Q18: How Similar Is LLM "Thinking" to Biological Brains?

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Learning objectives

- Compare transformer computation with biological neural computation at the mechanism level.
- Identify what the "brain" analogy gets right and where it misleads.
- Assess, soberly, whether building AI from brain structure is a promising path.

Companion video:

- *AI vs Human Thinking: How LLMs Really Work* — IBM Technology (12:11). https://www.youtube.com/watch?v=-ovM0daP6bw

1 The question, made precise

Student responses #06 and #11 ask how similar LLM "thinking" is to human thinking, how similar AI systems are to biological brains, and whether we could build AI from brain structure. To answer without hand-waving we must separate three claims: (1) behavioral similarity, (2) representational similarity, and (3) mechanistic similarity. They are often conflated, and they have very different evidence.

2 Where the analogy holds

- **Distributed representations.** Both brains and nets encode concepts across many units rather than one-unit-per-concept, and both show *superposition* (more features than neurons) [1].
- **Hierarchy of features.** Vision models and visual cortex both build edge → texture → part → object hierarchies, and model activations can *predict* neural responses in sensory cortex above chance [4].
- **Learning from experience.** Both adapt connection strengths from data rather than being fully hand-wired.

3 Where it breaks

- **Learning rule.** Backpropagation requires a global, symmetric backward pass that biological synapses do not plausibly implement; the brain's credit-assignment mechanism is unknown and likely local and different [3].

- **Substrate and dynamics.** Neurons are spiking, stochastic, continuous-time, energy-frugal (∼20 W whole brain); transformers are clocked, deterministic-ish, and enormously power-hungry.
- **Objective.** An LLM is trained to predict the next token on text; a brain is shaped by evolution and embodied, multi-modal, reward-driven interaction with a physical world. Same word ("learning"), very different pressures.
- **Architecture.** Attention over a fixed context window is not how cortex routes information; there is no biological KV cache.

> **Caution.** Behavioral similarity (an LLM "reasons" like a person on a benchmark) does not imply mechanistic similarity (it does so *the way* a person does). The strongest current evidence is representational (model activations predicting neural data in *sensory* systems); evidence for shared mechanism in high-level reasoning is weak.

4 A worked comparison: one task, two systems

Consider recognizing a handwritten digit. In a convolutional network, the pixel grid passes through stacked filters; early filters respond to oriented edges, later units to strokes and whole digits, and a final softmax commits to a label—one deterministic forward sweep, then a backward sweep at training time to adjust every weight by the global loss gradient. In the primate visual system the same stimulus drives a cascade from retina through V1 (edge-selective simple cells), V2/V4 (contours, shapes), to inferotemporal cortex (object identity)—superficially the same hierarchy, which is exactly why models trained only to classify end up *predicting* the firing of these areas [4]. But the differences are not cosmetic: the cortical cascade is recurrent (massive feedback between stages), runs in continuous time with spikes, adapts through local plasticity rules rather than a global backward pass, and is embedded in a body that moves its eyes to gather the input. The lesson for students: the *representations* converge because both systems are solving the same statistical problem under a hierarchy prior, while the *mechanisms* that build those representations remain distinct. Convergent representations are evidence about the problem, not proof of shared machinery.

5 Why the distinction matters practically

This is not idle philosophy. If LLM and brain mechanisms differ, then (a) we cannot assume AI will fail or succeed where humans do—its error profile is its own (jagged capabilities, module Q10); (b) neuroscience findings transfer to AI only as loose priors, not blueprints, so "it's how the brain does it" is a weak argument for a design choice; and (c) safety and interpretability must study the model's *actual* computation rather than reasoning by human analogy. Treating the analogy as literal is a recurring source of both over- and under-estimation of AI.

6 Could we build AI from brain structure?

This is the neuromorphic / brain-inspired program. Honest status: biologically detailed models (spiking networks, cortical microcircuits) have *not* matched task performance of mainstream deep learning, while the ideas that *did* transfer were abstract (distributed representations, convolution loosely inspired by receptive fields, attention loosely inspired by selective attention), not literal wiring diagrams [2]. The useful lesson flows both ways: deep nets are now a leading *model* of sensory cortex, and neuroscience occasionally suggests priors. But "copy the connectome and run it" is not a demonstrated route to capability today.

7 Summary

LLMs and brains share distributed, hierarchical, experience-shaped representations—enough that models predict sensory neural activity. They differ sharply in learning rule, substrate, objective, and architecture. Behavioral resemblance is not mechanistic identity. Brain-inspired AI has contributed *abstractions*, not literal blueprints; building AI from brain structure remains a research aspiration, not a proven method.

References

[1] Nelson Elhage, Tristan Hume, Catherine Olsson, et al. Toy models of superposition. *Transformer Circuits Thread*, 2022.

[2] Demis Hassabis, Dharshan Kumaran, Christopher Summerfield, and Matthew Botvinick. Neuroscience-inspired artificial intelligence. *Neuron*, 95, 2017.

[3] Timothy P. Lillicrap, Adam Santoro, Luke Marris, Colin J. Akerman, and Geoffrey Hinton. Backpropagation and the brain. *Nature Reviews Neuroscience*, 21, 2020.

[4] Daniel L. K. Yamins, Ha Hong, Charles F. Cadieu, Ethan A. Solomon, Darren Seibert, and James J. DiCarlo. Performance-optimized hierarchical models predict neural responses in higher visual cortex. *PNAS*, 111, 2014.